K-means clustering. K-means partitions n observations into k clusters by repeatedly assigning each point to the nearest centroid and then recomputing every centroid as the mean of its assigned points. The algorithm minimises the within-cluster sum of squares and stops when assignments no longer change. The number of clusters k must be chosen in advance. The elbow method plots the within-cluster sum of squares against k and looks for the point where the improvement flattens. The silhouette score compares how close a point is to its own cluster with how close it is to the nearest other cluster, with values near 1 indicating well separated clusters. K-means is sensitive to the initial centroids, so it is usually run several times.

Logistic regression and classification metrics. Logistic regression predicts the probability of a binary outcome by applying the sigmoid function to a linear combination of predictors. A decision threshold, commonly 0.5, converts the probability into a class label. Precision is the fraction of predicted positives that are truly positive. Recall is the fraction of true positives that the model finds. The F1 score is the harmonic mean of precision and recall. Lowering the threshold usually raises recall and lowers precision. Accuracy can be misleading when classes are imbalanced.

Decision trees and random forests. A decision tree splits the data using feature thresholds chosen to reduce impurity; the Gini impurity of a node is one minus the sum of squared class proportions. Deep trees overfit easily. A random forest trains many trees on bootstrap samples of the data and considers only a random subset of features at each split, then averages their predictions. This bagging approach reduces variance without greatly increasing bias. The out-of-bag error provides a built-in estimate of test error.